

Probabilistic Forecasting of Winter Weather Impacts on Solar Power Production

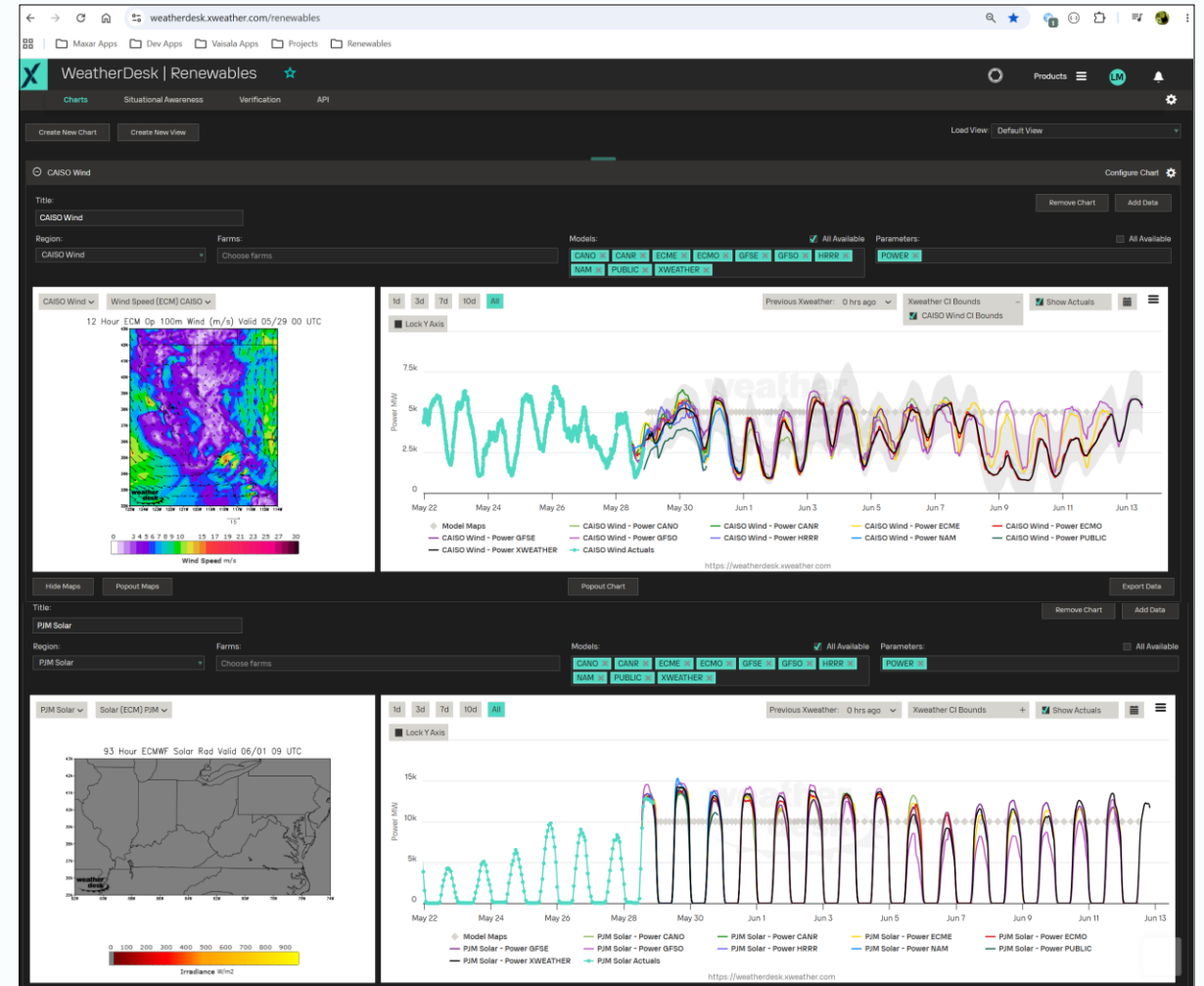
Leigh Munchak, Vaisala Xweather



Photo Courtesy of Sandia National Laboratories

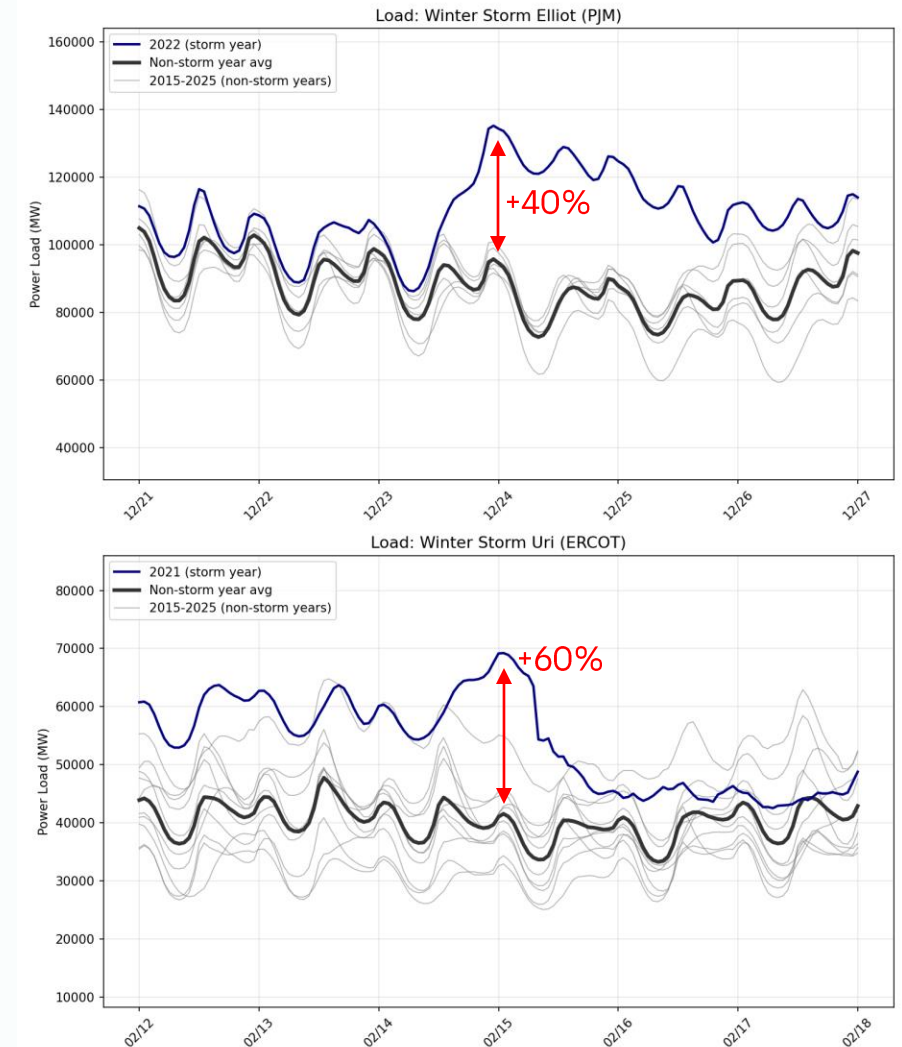
Renewable energy (RE) forecasting at Vaisala

- Vaisala's WeatherDesk Renewables Forecasting Team = Legacy 3Tier + Former Maxar
- WeatherDesk offers solar & wind forecasts for every RTO in the US
 - Forecasts for region total, any subregional totals, and all solar and wind farms within an RTO ("component farms")
- We also forecast wind and solar power generation for bespoke locations from clients ("site-specific farms")



Winter storms stress power supply & demand

- Winter storms pose unique challenges in balancing the grid
 - Recent(-ish) storms Uri (2021) & Elliot (2022) required load shedding events in some balancing areas
- Peak load significantly higher than 10-year average
- Decreased generation due to wind turbine icing, natural gas disruptions
- Expected tight grid conditions require highly accurate forecasts and some characterization of the risk (likelihood) and impact (severity)



Winter weather impacts on solar power production

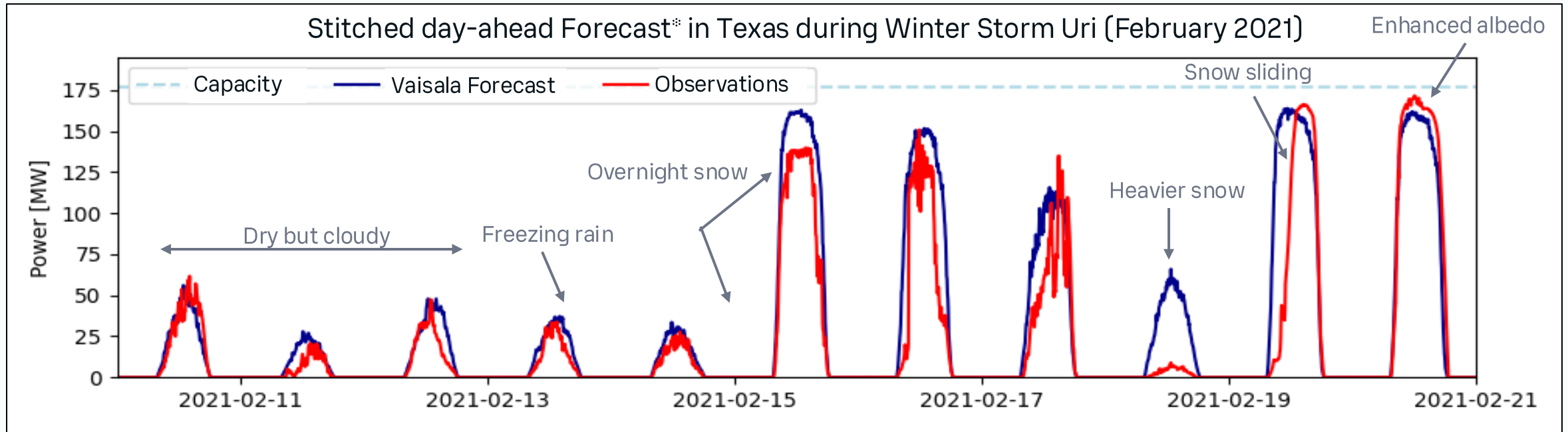


Snowfall – reduces irradiance on panels



Ice & freezing rain – damages infrastructure

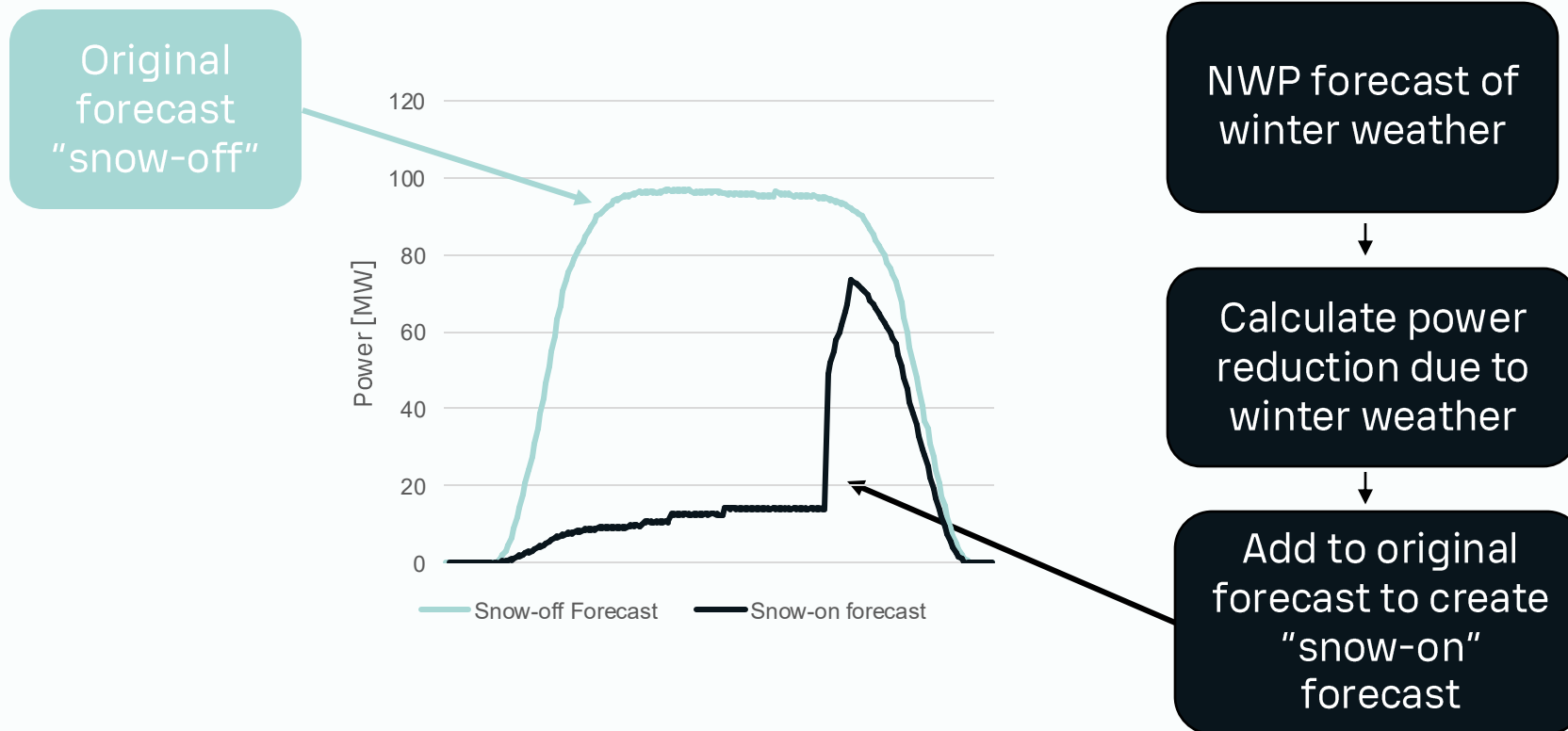
Example of a winter storm impacts on a solar forecast



*stitched day-ahead forecast means that this is a series of 11 forecasts, each issued in the morning and valid the next day (18-42 hours ahead)

Winter weather solar power forecasting methodology

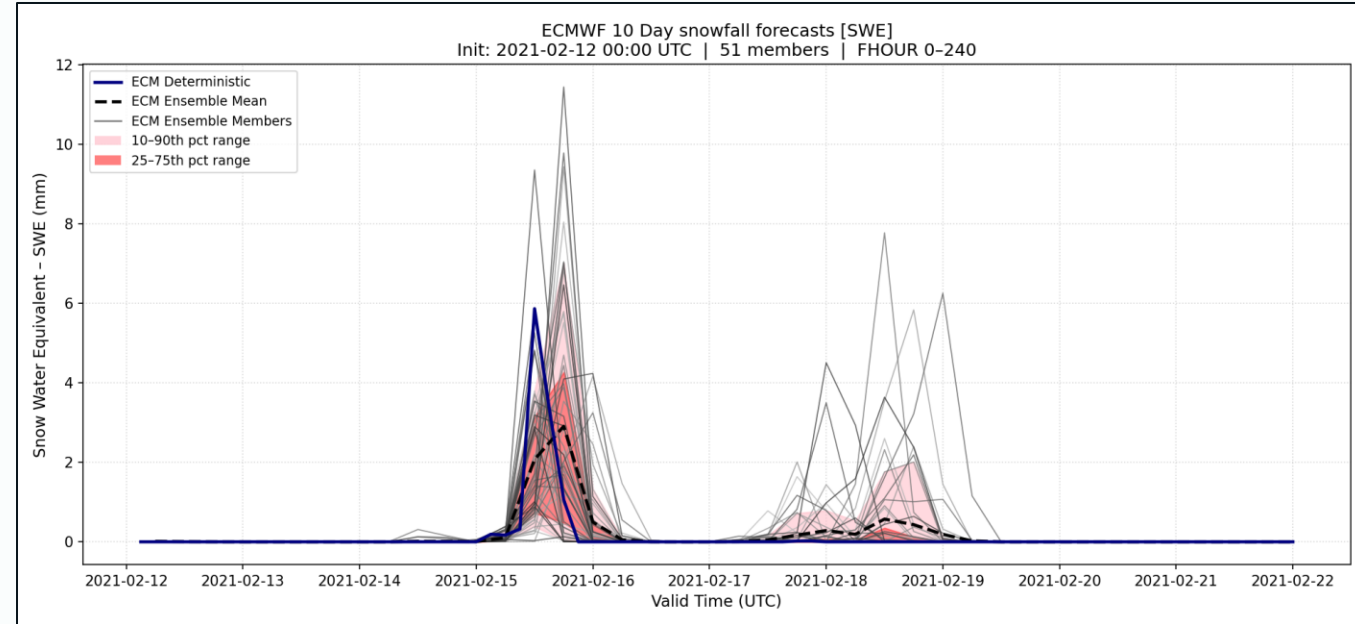
- Forecast power reduction from winter weather as a separate module, and then provide that as an additional forecast



Ensemble NWP Forecasts

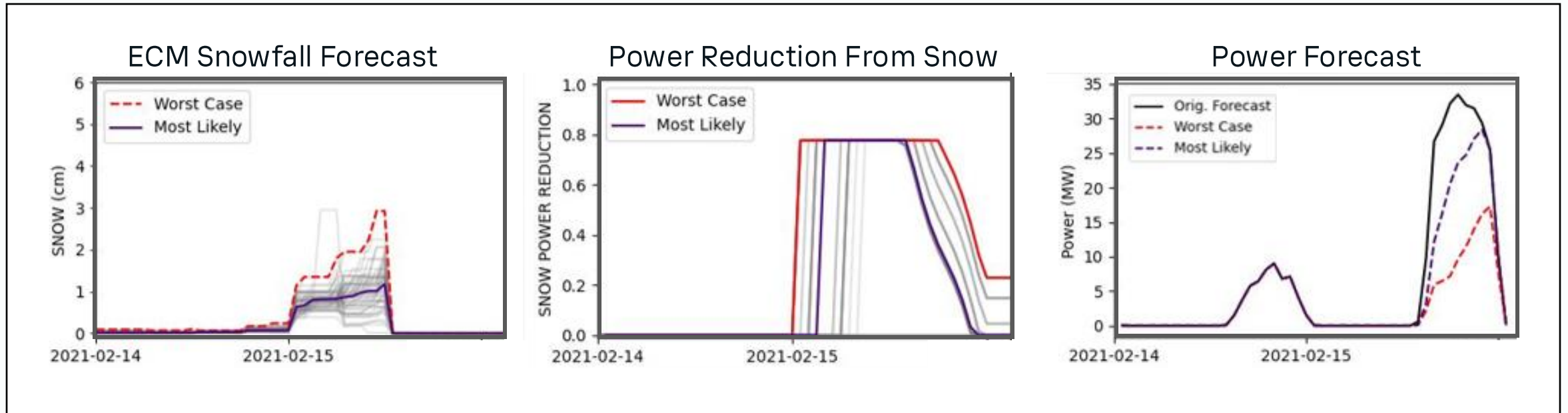


- Deterministic forecasts of winter weather didn't give enough information about risk – needed a probabilistic approach
- Decided ensemble NWP was most valuable for spread of winter weather possibilities
 - Decided on ECM ensemble for balance of update frequency (4x/day), forecast length (144/360 hours), accuracy, spread (50 ensemble members) and coverage (global)



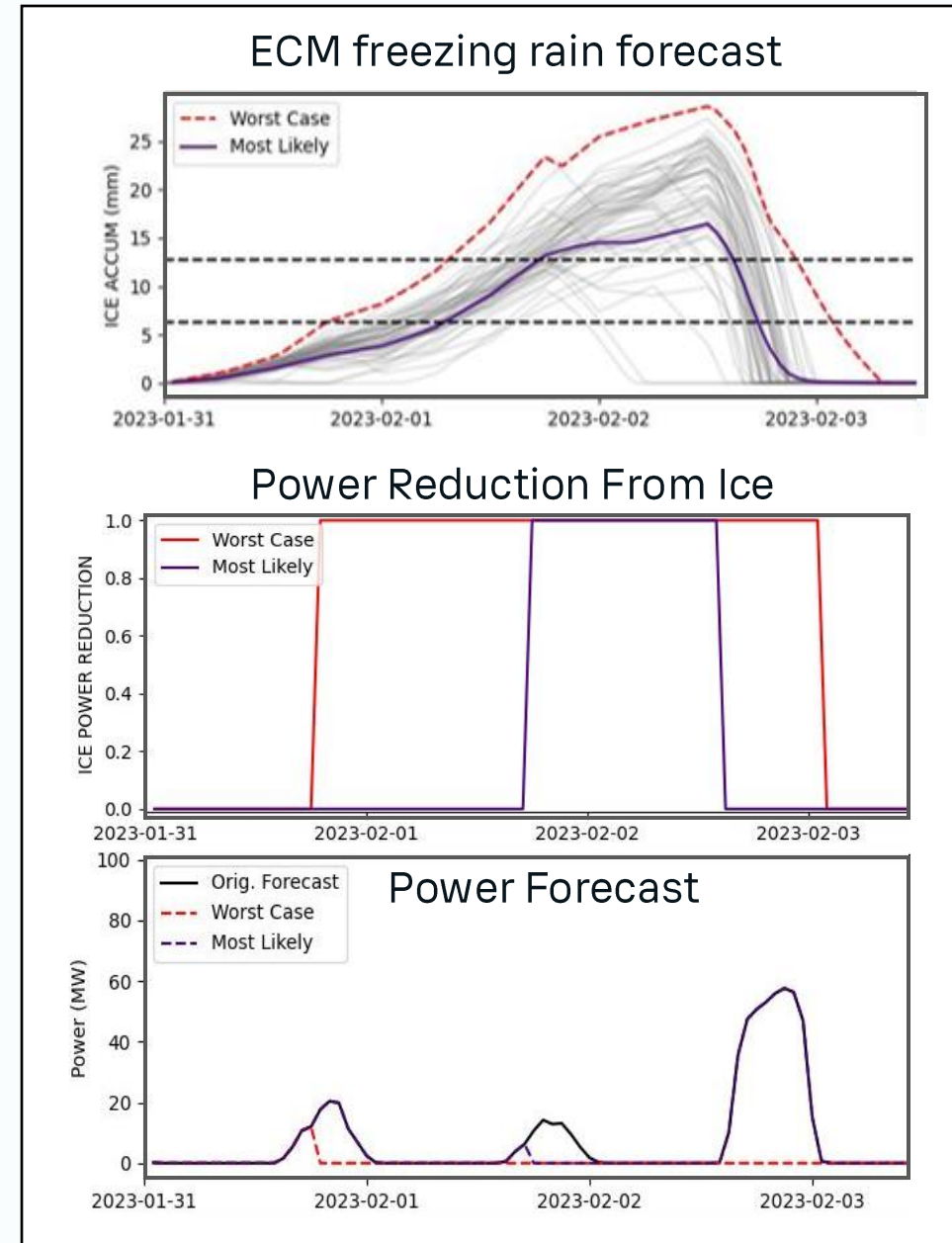
Snowfall Impacts

- Implemented a physically-based snow-soiling algorithm with snow sliding, along with additional correction algorithm incorporating power observations
 - Largely dependent on snowfall amount, surface temperature, and panel geometry
- Standard is to deliver “worst-case” and “most-likely” power reductions, but we internally keep the forecasts for all 51 members



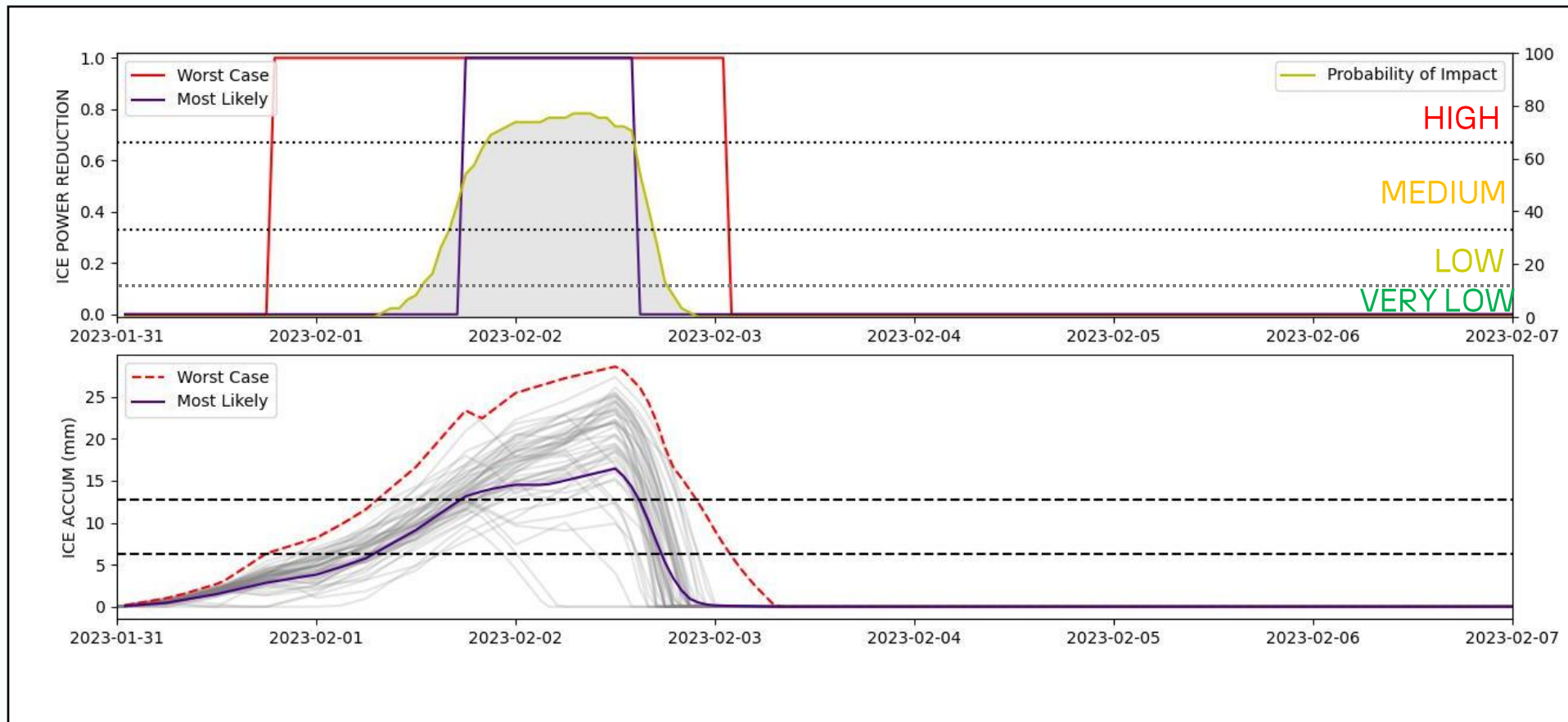
Icing Impacts

- Ice accumulation from freezing rain is transparent and therefore does not have a large impact on power production, but it can take equipment out of service.
- Since we cannot quantify a predictable impact, the most likely and worst-case power reductions from icing are modeled as either 0% or 100% of a site's operating capacity.



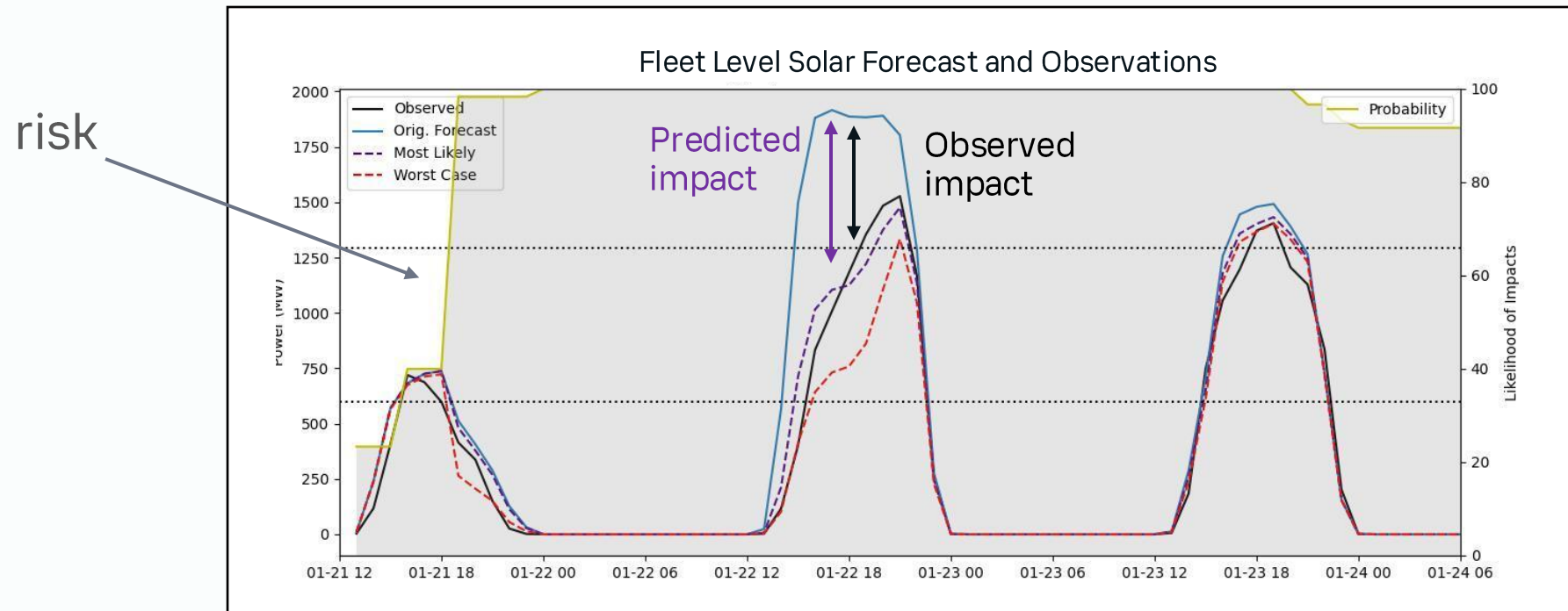
Quantification of risk

- Use the ensemble member power reduction forecasts from both snow and ice to calculate a "probability of impact"



Forecast Validation – Risk and Impact

- Validation is primarily via case studies – hard to bulk-validate extreme events because they are infrequent
- Day-ahead forecast predicts reduction of 25-50% power production due to snow, with 100% likelihood



Winter Weather Impact Alerting

- Automated emails are issued twice a day if significant winter weather impacts are expected
- Allows for pro-active fleet management

Fleet Level Winter Weather Impact Report

Event 1:

Summary of Impacts:

- Event Start: 2025-01-22 13:00:00 CT
- Event End: 2025-01-23 14:00:00 CT
- Likelihood of reduced solar generation during the event: 96.92%
- Category: HIGH

risk



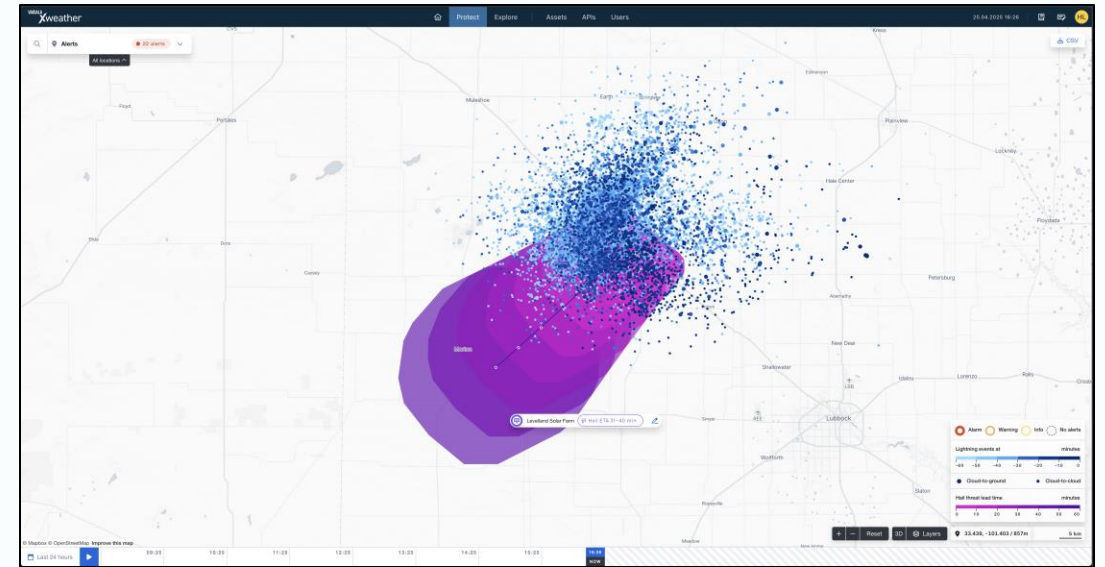
impact



	Extreme Wx Probability	Extreme Wx Probability Category	Event Most Likely Solar Generation Reduction (MW)	Event Worst-Case Solar Generation Reduction (MW)
County 1	0.00%	VLOW	0.00	0.00
County 2	0.00%	VLOW	0.00	0.00
County 3	9.29%	VLOW	0.00	-94.52
County 4	96.92%	HIGH	-333.50	-748.41
County 5	0.00%	VLOW	0.00	0.00
County 6	1.47%	VLOW	0.00	-66.48

Last thoughts on solar forecasting for extreme weather events

- Winter weather is a peril that makes sense to explicitly forecast impacts on power production
 - Larger scale, longer duration
- Other perils are better suited for a proactive alerting approach, without quantification
- “Forecasts possess no intrinsic value, they acquire value through their ability to influence decisions made by users of the forecast” (Murphy 1993)
 - Does our reliance on average MAE as the prime quantifier of forecast accuracy de-value extreme event forecasts?



0-60 minute hail alerting swath in Xweather Protect

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